

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S1 (S, FE) S2 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: MAT 102**Course Name: VECTOR CALCULUS, DIFFERENTIAL EQUATIONS AND TRANSFORMS (2019 SCHEME)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all Questions. Each question carries 3 Marks*

Marks

- 1 Show that the vector $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ is irrotational. (3)
- 2 Identify the velocity, acceleration and speed of a particle moving along the curve, $\vec{r} = (2 + 4t) \hat{i} + (1 - 3t) \hat{j} + (4 + 3t) \hat{k}$. (3)
- 3 Identify the outward flux of the vector field $F(x, y, z) = x \hat{i} + 2y \hat{j} + 3z \hat{k}$ across the unit sphere by using divergence theorem. (3)
- 4 Determine whether the vector field, $\vec{F}(x, y, z) = (y + z) \hat{i} - (xz^3) \hat{j} + (x^2 \sin y) \hat{k}$ is free of sources and sinks. If it is not, locate them. (3)
- 5 Find the general solution of $y'' - y = 0$. (3)
- 6 Show that e^{2x} , e^{3x} are linearly independent solutions of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$. Find its general solution. (3)
- 7 Find the Laplace Transform of $\cos^2 \omega t$. (3)
- 8 Find the inverse Laplace Transform of $\frac{2\pi}{(s+\pi)^3}$. (3)
- 9 Find the Fourier sine transform of $f(x) = e^{-x}$. (3)
- 10 Find the Fourier transform of $f(x) = \begin{cases} k & x < a \\ 0 & x > a \end{cases}$. (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module I**

- 11 (a) Find $\int_C (2y + 3)dx + xz dy + (yz - x)dz$ along the path $x = 2t$, $y = t$, $z = t$ from $(0, 0, 0)$ to $(2, 1, 1)$. (7)

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- (b) Examine whether $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ is a conservative vector field. If so, find the potential function. (7)

OR

- 12 (a) Find the directional derivative of $f(x, y, z) = x^2y - yz^3 + z$ at the point $(1, -2, 0)$ in the direction of the vector, $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$. (7)
- (b) A particle moves in three-dimensional space with velocity $\vec{v}(t) = \hat{i} + t\hat{j} + t^2\hat{k}$ where t is the time variable. Find the position vector of the particle when $t = 1$ given that the particle is at $(-1, 2, 4)$ when $t = 0$. (7)

Module II

- 13 (a) Find $\int_C [(\cos x \sin y - xy)dx + \sin x \cos y dy]$ by Green's theorem, where C is the unit circle $x^2 + y^2 = 1$. (7)
- (b) Use Stoke's theorem to calculate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = xy\hat{i} + yz\hat{j} + xz\hat{k}$ and C is the triangle in the plane $x + y + z = 2$ with vertices $(2, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 2)$ with counter-clockwise orientation looking from the first octant towards the origin. (7)

OR

- 14 (a) Using divergence Theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F}(x, y, z) = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$ and σ is the surface of the cylindrical solid bounded by $x^2 + y^2 = 4$, $z = 0$ and $z = 4$. (7)
- (b) Use Stoke's theorem to evaluate $\int_C e^x dx + 2y dy - dz$ where C is the curve $x^2 + y^2 = 4$, $z = 2$. (7)

Module III

- 15 (a) Solve $y'' + 4y = \tan 2x$ by the method of variation of parameters. (7)
- (b) Find the particular integral of $(D^2 + 4D + 1)y = e^x \sin 3x$. (7)

OR

- 16 (a) Solve the initial value problem $(x^2 D^2 + xD + 1)y = 0$, $y(1) = 1$, $y'(1) = 1$ (7)
- (b) Solve $y'' - 2y' + 5y = x^2$. (7)

Module IV

- 17 (a) Find (i) $L(t^3 e^{-2t})$ (ii) $L^{-1}\left(\frac{4s+32}{s^2-16}\right)$ (7)

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- (b) Solve the initial value problem using Laplace transforms:

$$y'' - 2y' - 8y = 0, y(0) = 3, y'(0) = 6 \quad (7)$$

OR

- 18 (a) Find the inverse Laplace Transform of $\frac{5}{(s^2+1)(s^2+25)}$ using convolution Theorem (7)

Solve the IVP

- (b) $y'' + 3y' + 2y = \delta(t-1), y(0) = 0, y'(0) = 0$ (7)

Module V

- 19 (a) Represent $f(x) = \begin{cases} x^2 & 0 < x < 1 \\ 0 & x > 1 \end{cases}$ as a Fourier Cosine Integral (7)

- (b) Determine the Fourier Transform of $f(x) = \begin{cases} e^{kx} & \text{if } x < 0 \\ 0 & \text{otherwise} \end{cases}$ where $k > 0$ (7)

OR

- 20 (a) Using Fourier integral representation, show that (7)

$$\int_0^\infty \frac{\sin w - w \cos w}{w^2} \sin xw \, dw = \begin{cases} \frac{\pi}{2}x, & \text{if } 0 < x < 1 \\ \frac{\pi}{4}, & \text{if } x = 1 \\ 0, & \text{if } x > 1 \end{cases}$$

- (b) Find the Fourier integral representation of the function (7)

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$
